

Dynamical Systems That Sort Lists, Diagonalize Matrices, and Solve Linear Programming Problems

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ABSTRACT

We study a class of smooth, continuous-time dynamical systems defined on the space of real symmetric matrices. The principal equation of interest is the double-bracket commutator flow $dH/dt = [H, [H, N]]$, where $H(t)$ and N are symmetric $n \times n$ matrices and $[A, B] = AB - BA$ is the standard matrix commutator (Lie bracket). We prove that this differential equation defines an isospectral gradient flow on the adjoint orbit of the orthogonal Lie group $O(n)$. When N is chosen as a fixed diagonal matrix with distinct, ordered entries ($N = \text{diag}(\mu_1, \dots, \mu_n)$ with $\mu_1 < \dots < \mu_n$), the steady-state solution $H(\infty)$ is diagonal, with its eigenvalues arranged in strictly ascending order. Consequently, the flow sorts arbitrary lists of numbers, computes matrix eigenvalues, and solves linear programming problems in continuous physical time.

1. Introduction and Motivation

Traditional algorithms in computer science treat sorting as an inherently discrete, combinatorial task requiring sequential comparisons and swaps (such as Quicksort, Heapsort, or Mergesort). However, in analog computing, physics, and control theory, optimization problems often admit natural representations as continuous dynamical systems that converge to the desired solution as $t \rightarrow \infty$.

In this paper, we establish that the basic algorithmic task of sorting a list of real numbers can be realized as the natural evolution of a continuous, autonomous differential equation on a smooth manifold.

The underlying mathematics connects Riemannian geometry, Lie algebras, Hamiltonian mechanics, and the classical Rearrangement Inequality of Hardy, Littlewood, and Polya.

2. Lie Bracket Formulation & Geometric Structure

Let $S(n)$ denote the vector space of $n \times n$ real symmetric matrices, and let $\mathfrak{so}(n)$ denote the Lie algebra of $n \times n$ skew-symmetric matrices: $\mathfrak{so}(n) = \{ \Omega \in \mathbb{R}^{n \times n} : \Omega^T = -\Omega \}$.

Equip $S(n)$ with the canonical Frobenius inner product: $\langle A, B \rangle = \text{Tr}(AB)$.

Given an initial symmetric matrix H_0 in $S(n)$, consider the adjoint orbit $O(H_0)$ under the action of the orthogonal group $O(n)$:

$$O(H_0) = \{ \Theta H_0 \Theta^T : \Theta \in O(n) \}$$

Every matrix in $O(H_0)$ has the exact same set of eigenvalues as H_0 . Thus, motion along $O(H_0)$ is strictly isospectral.

3. Main Theorems and Proofs

Theorem 1 (Isospectral Flow Property): Let $H(t)$ satisfy $dH/dt = [H, [H, N]]$ with $H(0) = H_0$ in $S(n)$ and N in $S(n)$. Then for all t in $\mathbb{R}$: (1) $H(t)$ is symmetric, (2) The eigenvalues of $H(t)$ are strictly independent of time t , and (3) Any stationary point H^* satisfies $[H^*, N] = 0$.

Proof: Let $\Omega(t) = [N, H(t)]$. Since H and N are symmetric: $\Omega^T = (NH - HN)^T = HN - NH = -[N, H] = -\Omega$. Hence $\Omega(t)$ in $\mathfrak{so}(n)$ for all t . The ODE can be written as $dH/dt = [\Omega, H]$. Consider the matrix ODE on $O(n)$: $d\Theta/dt = \Omega(t)\Theta(t)$ with $\Theta(0) = I$. Then $H(t) = \Theta(t)H_0\Theta(t)^T$. Because $H(t)$ is related to H_0 by an orthogonal similarity transformation, its eigenvalues are identically preserved for all t . Q.E.D.

Theorem 2 (Gradient Flow on the Adjoint Orbit): The double-bracket flow $dH/dt = [H, [H, N]]$ is the steepest ascent gradient flow of the linear functional $\Phi(H) = \text{Tr}(HN)$ on the Riemannian manifold $O(H_0)$ equipped with the normal metric induced by the Lie algebra.

Proof: Let $\delta H = [\Omega, H]$ in $T_H O(H_0)$. The directional derivative is $d\Phi(H)(\delta H) = \text{Tr}((\delta H)N) = \text{Tr}([\Omega, H]N) = \text{Tr}(\Omega[H, N])$. Under the inner product $\langle A, B \rangle = -1/2 \text{Tr}(AB)$ on $\mathfrak{so}(n)$, the gradient corresponds to $\Omega^* = [N, H]$. Projecting to the tangent space yields $\text{grad } \Phi(H) = [[N, H], H] = [H, [H, N]]$. Q.E.D.

4. Asymptotic Sorting via the Rearrangement Inequality

Theorem 3 (Global Attractor and Sorting Property): Suppose $N = \text{diag}(\mu_1, \dots, \mu_n)$ has distinct ordered eigenvalues $\mu_1 < \mu_2 < \dots < \mu_n$. Let H_0 have distinct eigenvalues $\lambda_1 < \dots < \lambda_n$. Then:

1. The critical points of $\Phi(H)$ on $O(H_0)$ consist of all diagonal matrices whose entries are permutations of $\{\lambda_1, \dots, \lambda_n\}$. There are exactly $n!$ isolated critical points.
2. The global maximum of $\Phi(H)$ is uniquely attained at $H^* = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$.
3. The only asymptotically stable equilibrium point of the flow is H^* . For almost all initial conditions H_0 , $H(t)$ converges as $t \rightarrow +\infty$ to H^* , sorting the eigenvalues in strictly increasing order.

Proof: At equilibrium, $[H^*, N] = 0$. Since N has distinct diagonal entries, any symmetric matrix commuting with N must be diagonal: $H^* = \text{diag}(\lambda_{\pi(1)}, \dots, \lambda_{\pi(n)})$. By the classical Rearrangement Inequality (Hardy, Littlewood, and Polya, 1934), $\sum_{i=1}^n \lambda_{\pi(i)} \mu_i$ attains its unique global maximum if and only if π is the identity permutation.

The Hessian at a critical point is $\delta^2 \Phi = -(\lambda_{\pi(i)} - \lambda_{\pi(j)})(\mu_i - \mu_j)$. For H^* to be a local maximum, the Hessian must be negative definite, requiring $\lambda_{\pi(i)} < \lambda_{\pi(j)}$ for all $i < j$. This uniquely selects the sorted order! All other $(n! - 1)$ critical points are unstable saddle points. Q.E.D.

5. Analog and Physical Implementation

In physical hardware (analog circuits, optical crossbars, and quantum simulators), matrix commutators can be executed in continuous physical time without sequential clock cycles.

Practical physical implementations are governed by signal-to-noise ratio (SNR), analog readout time, and circuit dissipation, offering a compelling alternative to sequential Turing machines.